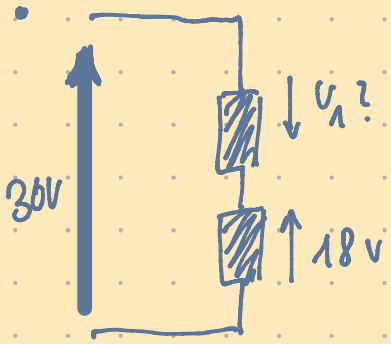
Revision Electricite

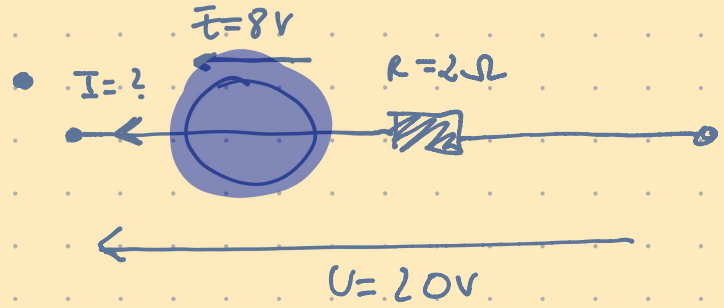
Partie A

Exo 1



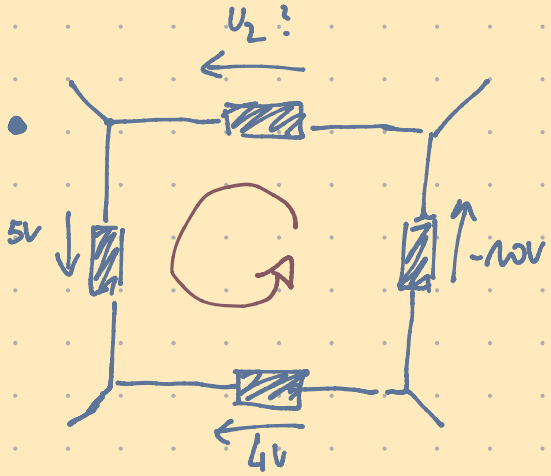
$$30 = -U_1 + 18$$

$$U_1 = 18 - 30$$
$$= -12V$$



Loi d'additivité :

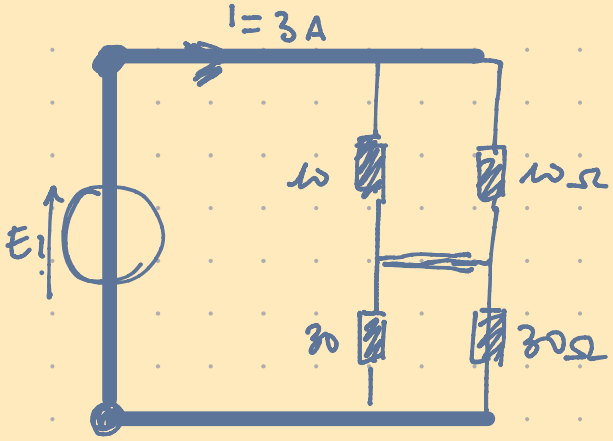
$$U = E - RI \quad \Rightarrow \quad I = \frac{E - U}{R}$$
$$= \frac{8-20}{2}$$
$$= -6A$$



$$U_2 + 5 - 4 - 10 = 0$$

$$U_2 - 9 = 0 \Rightarrow U_2 = 9V$$

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$$R_{eq} = (10 \,//\, 10) + (30 \,//\, 30)$$

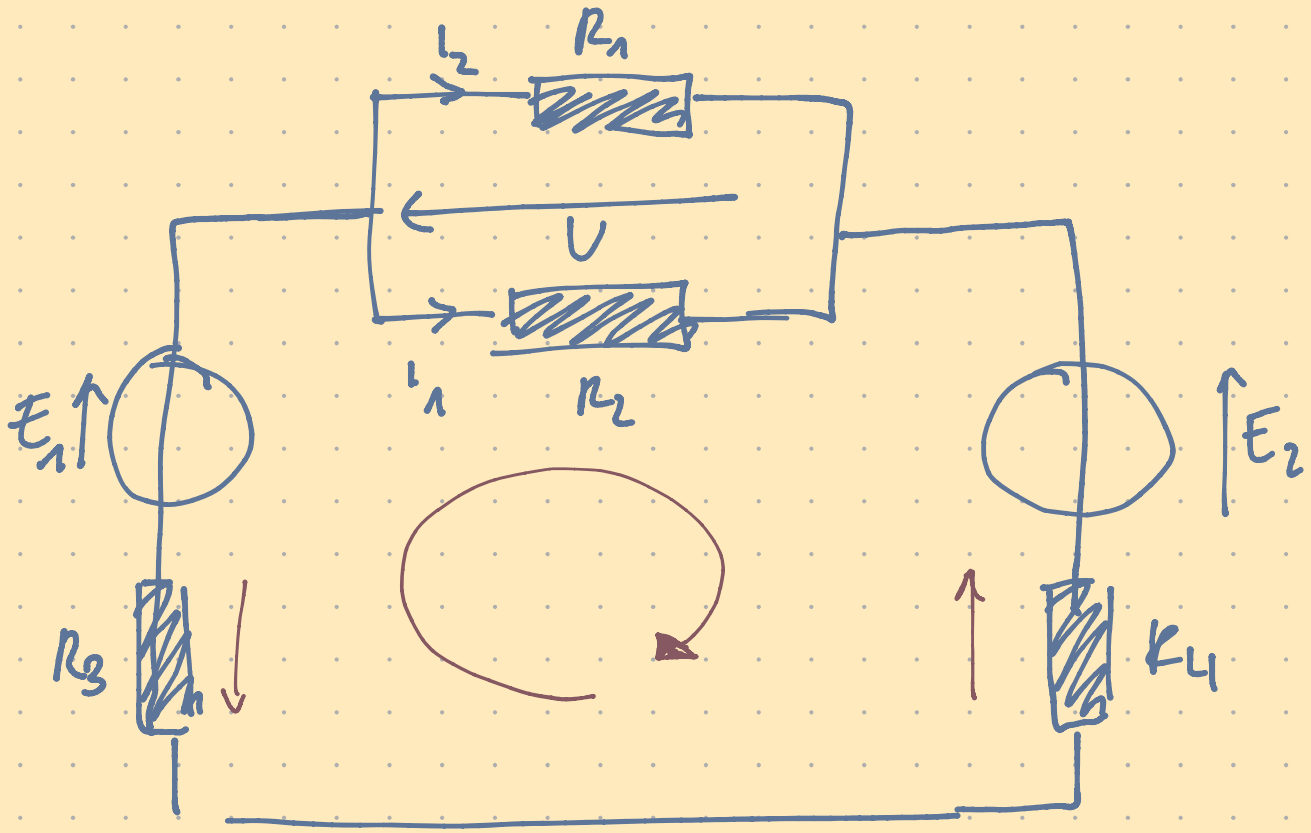
$$= \frac{10 \times 10}{10+10} + \frac{30 \times 30}{30+30}$$

$$= \frac{100}{20} + \frac{900}{60}$$

$$= \frac{10}{2} + \frac{30}{2} = 20\Omega$$

$$E = R_1 I$$
$$= 20 \times 3$$
$$= 60V$$

Exo 2



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$E_2 = 30V \quad R_1 = 50\Omega \quad R_2 = 75\Omega \quad R_3 = 4\Omega \quad R_4 = 6\Omega$

$U = 15V$

1) $U = R_1 I_1 = R_2 I_2$

$\Rightarrow I_1 = \frac{U}{R_1} = \frac{15}{50} = 0{,}3\,A \qquad I_2 = \frac{U}{R_2} = \frac{15}{75} = 0{,}2\,A$

$I = I_1 + I_2 = 0{,}3 + 0{,}2 = 0{,}5A$

2) $E_1 - U - E_2 - (R_3 + R_4) \cdot i = 0$

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